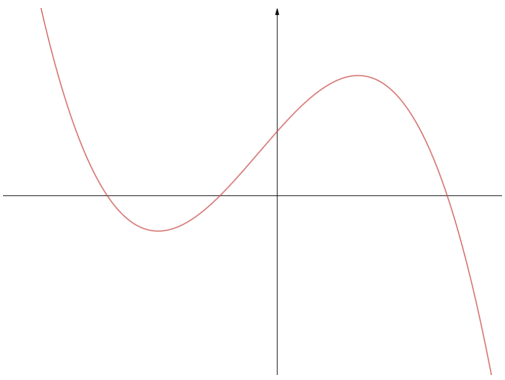
Question	Answer	Marks	Partial Marks
8(b)	$2x^2 + 11x - 21$ [= 0]	**M2**	or **M1** for $8 - 5x = 2x^2 + 6x - 13$ oe or better
	$(2x - 3)(x + 7)$ [= 0] oe or $\frac{-11 \pm \sqrt{11^2 - 4 \times 2 \times (-21)}}{2 \times 2}$ or $-\frac{11}{4} \pm \sqrt{\frac{21}{2} + \left(\frac{11}{4}\right)^2}$ oe	**M2**	Allow correct method to solve *their* quadratic equation e.g. formula, complete the square but not for $2x^2 + 6x - 13$ **M1** FT *their* equation for $2x(x + 7) - 3(x + 7)$ [= 0] or $x(2x - 3) + 7(2x - 3)$ [= 0] or $(2x + a)(x + b)$ [= 0] where $ab = -21$ or $2b + a = 11$ OR **M1** for $\sqrt{11^2 - 4 \times 2 \times -21}$ or for $\frac{-11 + \sqrt{k}}{2 \times 2}$ or $\frac{-11 - \sqrt{k}}{2 \times 2}$ OR **M1** for $\left(x + \frac{11}{4}\right)^2$
	$\left(\frac{3}{2}, \frac{1}{2}\right)$ and (–7, 43)	**B2**	**B1** for one correct pair or for 2 correct x-values or 2 correct y-values
9(a)	Correct sketch of negative cubic crossing the x-axis at –3, –1 and 3 and crossing the y-axis at 9 	**4**	**B1** for any negative cubic shape with two turning points **B2** for three intercepts only with x-axis labelled at – 3, –1 and 3 or **B1** for one or two correctly labelled x-intercepts **B1** for intercept with y-axis labelled at 9 If no graph drawn, **SC1** for all four intercepts labelled on axes.
9(b)(i)	$3 - x + 3x - x^2$ or better or $3 + x + 3x + x^2$ or better or 9 [$- 3x + 3x$] $- x^2$	**M1**	At least 3 of the four terms correct or for the correct expansion of all three brackets with all 8 terms correct
	Correct completion to [y =] $9 + 9x - x^2 - x^3$	**A1**	with no errors or omissions seen

Question	Answer	Marks	Partial Marks
9(b)(ii)	$9 - 2x - 3x^2 = 0$ oe	**B3**	**B2** for $9 - 2x - 3x^2$ or **B1** for two correct terms **M1** for *their* derivative = 0 or stating $\frac{dy}{dx} = 0$
	$\frac{--2 \pm \sqrt{(-2)^2 - 4 \times -3 \times 9}}{2 \times -3}$ oe OR $-\frac{1}{3} \pm \sqrt{\frac{9}{3} + \left(\frac{1}{3}\right)^2}$ oe	**B2**	FT *their* derivative **B1FT** for $\sqrt{(-2)^2 - 4(-3)(9)}$ or better or for $\frac{-(-2) + \sqrt{q}}{2 \times -3}$ or $\frac{-(-2) - \sqrt{q}}{2 \times -3}$ OR **B1** for $\left(x + \frac{1}{3}\right)^2$
	–2.10 and 1.43 final answer	**B2**	**B1** for each or for answers –2.1 or –2.097 … and 1.4 or 1.430 to 1.431 or **SC1** for –2.097... **and** 1.43[0] to 1.431 seen in working or for –1.43 **and** 2.10 as final answer
9(b)(iii)	[a =] – 6 [b =] 17	**3**	**B2** for either a correct or b correct or for [a =] –5.04 or –5.049 to –5.05 **and** [b =] 16.9… seen or **M1** for substitution of one of *their* solutions into $9 + 9x - x^2 - x^3$ oe or **SC1** for reversed answers, $a = 17$, $b = -6$
10(a)	20.8 or 20.76 to 20.79	**4**	**B3** for [BC =] 10.4 or 10.38 to 10.39… or $6\sqrt{3}$ oe or **M2** for $(2x)^2 + x^2 + 6^2 = 24^2$ oe or **M1** for $24^2 - 6^2$ oe or $x^2 + 6^2$ oe or $(2x)^2 + 6^2$ oe, or $x^2 + (2x)^2$ oe or **SC2** for final answer of $12\sqrt{5}$ or 26.8 or 26.83… OR **M3** for $x^2 + \left(\frac{x}{2}\right)^2 + 6^2 = 24^2$ oe or **M2** for $x^2 + \left(\frac{x}{2}\right)^2$ or **M1** for $x^2 + 6^2$ oe or $\left(\frac{x}{2}\right)^2 + 6^2$ oe or $24^2 - 6^2$ oe